

Family history of hematopoietic and other cancers in children with acute lymphoblastic leukemia

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Abstract

It has been difficult to show a genetic contribution for many cancers, resulting in the belief that rare cancers such as leukemia are sporadic. We carried out a population-based case-control study including 701 incident cases of acute lymphoblastic leukemia (ALL) aged 0–14 years and diagnosed between 1980 and 1998 in Québec (Canada); 701 healthy controls were matched on age and sex. Cancer family history in the child's relatives was reported by parents. A positive family history of hematopoietic malignancies among first- or second-degree relatives was associated with a slight increase of risk for childhood ALL (odds ratio 2.06; 95% confidence interval = 1.22–3.49); the excess was attributable to second-degree relatives, and observed for grandparents as well as for uncles/aunts. The risk of ALL was not increased in children with a family history of cancers other than hematopoietic cancers. The data suggest a modest familial contribution to childhood ALL. © 2004 International Society for Preventive Oncology. Published by Elsevier Ltd. All rights reserved.

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1. Introduction

It has been difficult to show an inherited susceptibility for many cancers, resulting in the belief that rare cancers such as leukemia are sporadic. However, there are some data on cancer in first-degree relatives (parents, siblings, and offspring) of all-age cancer patients [1], and on cancer in all-age offspring whose parents or siblings had a concordant cancer [2,3], suggesting a familial contribution to leukemia. Recently, Risch reanalyzed these results [4], and those from a study on twins [5], which had suggested that for many cancers, environmental factors were more important determinants than genetic factors. He concluded that, on the contrary, most cancers, even the rare ones such as leukemia, are mostly genetically determined. Whether children with leukemia, in particular those with acute lymphoblastic leukemia (ALL), have first- or second-degree relatives (grandparents, uncles and aunts) with an increased incidence of hematopoietic or other cancers is not well established as only a few studies have addressed this [6–10].

We carried out a population-based case-control study of risk factors for ALL [11–13]. This study provided the op-

portunity to evaluate whether family history of cancer contributes to the risk of ALL.

2. Materials and methods

2.1. Case ascertainment

The study had two phases. During the first phase, all cases of ALL diagnosed between 1980 and 1993 in the Province of Québec aged between 0 and 9 years of age were determined to be eligible. Cases were recruited from tertiary care centers designated by government policy to treat and hospitalize children with cancer in the province; tracing cases from these hospitals is equivalent to a population-based ascertainment. In the second phase of the study, spanning from 1994 to 1998, recruitment was similar but included cases aged 0–14 years. Because cancer care is covered under the universal health plan, we believe that a negligible number of children, if any, were treated outside the province.

A case was determined to have acute lymphoblastic leukemia (International Classification of Diseases, Ninth Revision (ICD-9) coding 204.0) on the basis of clinical and biological standard criteria by an oncologist in a tertiary care center. To identify cases in the first phase of the

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study, we used: (1) hospitalization data from the provincial government's computerized discharge data files; (2) hospitalization censuses from the respective hospitals (we checked all medical records with a relevant discharge diagnosis); (3) lists maintained by hematology–oncology laboratories of histological data for cases; and (4) at the largest pediatric center in the province, medical records, distinct from the hospital medical record, which are maintained in the oncology outpatient clinic. In the second phase of the study, carried out at the same centers, recruitment was done by the clinical research personnel maintaining a database with all diagnosed patients; they provided us with the list of all eligible ALL cases and of those who had accepted to be contacted by us.

2.2. Control ascertainment

In the first phase of the study, population-based controls (one per case) were matched on sex and age at the time (calendar date) of diagnosis and thus were concurrently selected. The population-based controls were chosen from family allowance files. The family allowance is a government stipend awarded to all families with children living legally in Canada. This source of data was the most complete census of children for the study years. According to the expected distribution of cases based on matching criteria, a list of 10 potential controls was randomly chosen from the lists. In the second phase of the study, we used the provincial universal health insurance file as a source for controls, which is an equivalently complete census of children that in addition includes more demographic information to trace families. We proceeded the same way to obtain potential controls who were also matched on sex and age at the time of the diagnosis.

2.3. Study participants

We classified as not eligible the children who were adopted, lived in foster families, whose family spoke neither French nor English, were not resident in Canada, or whose parents were both unavailable for interview. In the first phase of the study, we identified 510 eligible cases and interviewed 491 parents (96.3%); 588 eligible control parents were contacted and 493 (83.8%) participated. Seventy-four controls included in the study were the second control on the list of eligibles, nine were the third choice, and one was the fourth. The others were first choices on the list of eligibles. In the second phase of the study, initial recruitment of cases was done by the treating centers. The eligibility criteria were the same; 237 cases were eligible and 210 accepted the interview (88.6%); 227 controls were contacted and 210 were interviewed (92%). Sixteen controls were the second control on the list of eligibles and one was the third. The others were first choices among eligibles. Reasons for non-participation were a confidential telephone number, refusal, or the family could not be traced. In the end, 491 pairs from the first phase and 210 from the second

could be used in the analysis for a total of 701 cases and as many controls. We requested that the parents return a signed informed consent form for the interview.

2.4. Data collection

Soon after the anticipated reception of a letter introducing the general purpose to the study, trained interviewers contacted the parents to schedule an appointment for the interview, which was administered by telephone using a structured questionnaire. Questionnaires were reviewed as they came in and feedback was given regularly to interviewers. The questionnaire included information on potential confounding factors and on family history of cancer; mothers and fathers were interviewed separately about their family history, and only mothers reported on cancer in siblings of the index child. The questions were: "Was there ever a diagnosis of cancer or leukemia made for you, your mother, your father, your siblings, or your children?" "If yes, which member of the family was affected by it?" "If yes, specify the type(s) of cancer and the age at which the cancer was diagnosed". The following were the names used by parents to report cancers that we grouped under hematopoietic malignancies (ICD-9 codes 200–208): leukemia, lymphocytic leukemia, acute lymphoblastic leukemia, lymphoma, lymphosarcoma, myeloma, Hodgkin's disease and cancer of the blood (used twice, once by a control father for a sibling diagnosed at age 19 years and once by a case mother for her mother diagnosed at age 43 years). Other cancers were from the categories 140–199, and 210–239. Ninety-nine percent (99%) of case mothers and 98% of control mothers answered the maternal questionnaire; 86.6% of case fathers answered the paternal questionnaire, whereas 81.6% of control fathers did. An additional 9.9% of case fathers' questionnaire were answered by the mother, whereas this percentage was 13.6% among control fathers.

2.5. Data analysis

The association between family history exposure and risk of ALL was estimated using conditional logistic regression. Odds ratios (OR) and 95% confidence intervals (95% CI) were estimated adjusting for mother's age and education. We report history of cancer and of other diseases (explained below) for first-degree relatives of the index child (his or her parents and siblings), and for second-degree relatives (his or her grandparents, uncles and aunts).

3. Results

The case and control groups each included 302 girls and 399 boys. In 675 pairs out of 701, the case and the matched control had an age difference of less than 3 months; the difference was between 3 and 12 months or more for 26 pairs. There were small differences between cases and controls in

Table 1
Characteristics of cases of acute lymphoblastic leukemia and their controls

Factor	Controls (n and %)	Cases (n and %)
Mother's level of schooling		
College or university	233 (33.3)	224 (31.9)
Secondary school	442 (63.1)	444 (63.3)
None, primary	25 (3.5)	33 (4.7)
Family income at diagnosis ^a		
>40,000	251 (36.9)	258 (37.3)
10,000–39,000	395 (58.1)	399 (57.6)
<10,000	34 (5.0)	35 (5.0)
Maternal age ^b		
<35 years	663 (94.5)	641 (91.4)
>35 years	38 (5.4)	60 (8.5)

^a In Canadian dollars (information available for 680 controls and 692 cases).

^b At birth of index child.

the distribution of mother's age at birth of the child and mother's education (Table 1). Among the 701 pairs of parents interviewed, family history of cancer was available for 690 sets of case and control mothers and 684 sets of case and control fathers. At least one cancer was reported in 53.1% of the case families and in 49.3% of the control families. Thirty-four percent of case mothers and 30.8% of control mothers reported a cancer diagnosis for themselves, their parents or their siblings (i.e., their own first-degree relatives). Among fathers, these figures were 31 and 29%, respectively.

Cases had 792 siblings at the time of diagnosis whereas controls had 707. No siblings of study subjects were di-

agnosed with hematopoietic malignancies. Three parents were diagnosed in this category: two case mothers (one had lymphosarcoma, and one had Hodgkin's disease), and one control father (who had ALL). A total of 45 second-degree relatives among cases and 23 similar relatives among controls were diagnosed with a hematopoietic malignancy, mainly leukemia generally not further characterized by cell type. Only one case child had a multiply affected family where the mother and a sibling of the father, as well as a sibling of the mother, were reported to have leukemia.

As shown in Table 2, there was no excess of hematopoietic malignancies reported among first-degree relatives, but a slight excess for second-degree relatives, which was consistently observed for grandparents as well as for uncles/aunts. Defining a positive family history based on one or more hematopoietic malignancies among the child's siblings, parents, grandparents, uncles and aunts, the odds ratio associated with such a history was 2.06 (95% CI = 1.22–3.49). A family history of hematopoietic malignancies was associated with a stronger risk increase on the maternal side (OR = 3.63; 95% CI = 1.65–7.99) than on the paternal side (OR = 1.18; 95% CI = 0.56–2.47).

The risk of ALL was not increased for a family history of one or more cancers other than hematopoietic cancers (Table 3). Only one case's sibling was diagnosed with cancer (neuroblastoma). When considering specific malignancies, only breast cancer was reported more frequently in the case's family than in the control's; among second-degree relatives, there was an increased risk of borderline significance both

Table 2
Adjusted odds ratios^a and 95% confidence interval for family history of hematopoietic malignancies (ICD-9, 200–208)

	Cases number (%)	Controls number (%)	Ratio of discordant pairs	Adjusted OR (95% CI)
First-degree relatives	2 (0.3)	1 (0.1)	2/1	1.80 (0.16–20.16)
Index's siblings	0	0	–	–
Mother	2 (0.3)	0	2/0	–
Father	0	1 (0.1)	0/1	–
Second-degree relatives	43 (6.2)	23 (3.3)	41/20	2.07 (1.21–3.55)
Grandparents	29 (4.2)	17 (2.5)	28/15	1.94 (1.01–3.59)
Uncles/aunts	15 (2.2)	6 (0.9)	15/6	2.45 (0.95–6.35)

^a Adjusted for maternal age and level of schooling.

Table 3
Adjusted odds ratios^a and 95% confidence interval for family history of cancer

Type of cancer (ICD-9)	First-degree relatives ^b		Second-degree relatives ^c	
	Ratio of discordant pairs	Adjusted OR (95% CI)	Ratio of discordant pairs	Adjusted OR (95% CI)
All cancers excluding hematopoietic malignancies	19/15	1.16 (0.59–2.31)	176/155	1.11 (0.90–1.38)
Breast (174)	5/2	1.99 (0.38–10.46)	74/49	1.48 (1.03–2.13)
Digestive (150–159)	1/2	0.42 (0.04–4.70)	91/81	1.09 (0.81–1.48)
Lung (162)	0/0	–	72/68	1.06 (0.76–1.47)
Genitourinary (179–189)	0/1	–	48/41	1.19 (0.78–1.80)

^a Adjusted for maternal age and level of schooling.

^b Includes parents and siblings of the index child.

^c Includes grandparents, uncles and aunts.

Table 4

Adjusted odds ratios^a and 95% confidence interval for family history of diabetes or asthma

	Cases number (%)	Controls number (%)	Ratio of discordant pairs	Adjusted OR (95% CI)
Diabetes				
First-degree relatives ^b	21 (3.1)	20 (2.9)	19/17	1.06 (0.55–2.07)
Second-degree relatives ^c	263 (38.5)	286 (41.6)	142/166	0.85 (0.77–1.14)
Asthma				
First-degree relatives ^c	153 (22.3)	127 (18.5)	123/98	1.26 (0.96–1.64)
Second-degree relatives ^c	207 (30.1)	217 (31.6)	151/160	0.94 (0.77–1.15)

^a Adjusted for maternal age and level of schooling.^b Includes parents and siblings of the index child.^c Includes grandparents, uncles and aunts.

for the grandmothers (OR = 1.4; 95% CI = 0.94–2.07) and the aunts (OR = 2.05; 95% CI = 0.97–4.36).

Table 4 shows results for family history of diabetes and asthma. No differences were observed when reporting a history of diabetes or asthma for first- or second-degree relatives. Family history of Crohn's disease was not different between cases and controls (data not shown).

4. Discussion

Overall, hematopoietic malignancies were reported more frequently among second-degree relatives of cases than of controls, resulting in a moderate increase in the risk of ALL. Risk was apparently not increased for the child's parents or siblings. The risk of ALL was not increased for a family history of one or more other cancers; there was an indication that breast cancer was slightly increased in the families of cases.

Using data from the Danish Cancer Registry, Olsen et al. [6] studied approximately 2000 leukemia cases below the age of 15 years, and found no excess of cancer in their parents. Winther et al. [7] did a similar study in the Nordic countries determining the risk of cancer in siblings for over 6000 cases of leukemia diagnosed before the age of 20 years. No increase in risk was observed. Three previous case-control studies from the US, France, and Moscow including childhood ALL cases measured the risk associated with family history of cancer as reported by parents. The earliest study included 203 cases of infant ALL and reported a cancer OR of 1.1 (95% CI = 0.8–1.7) for parents, and a similar OR for second-degree relatives [8]; the OR for hematopoietic cancers was 1.4 (95% CI = 0.7–3.0). The study from Moscow [9] included 199 cases of leukemia and reports an OR of 1.5 (95% CI = 1.04–2.1) for all cancers when including parents and grandparents. Finally, the French study [10] included 243 cases of childhood ALL and reported no increase in risk for hematologic malignancies for first-degree relatives, but an OR of 2.2 (95% CI = 0.8–6.0) for second-degree relatives. This study also reported an increased risk for solid tumors.

Study limitations and potential biases must be addressed. In our study, size of the sibship for cases and controls was

measured, and size is not an issue for parents and grandparents. However, while we obtained information on cancer diagnosis among parents' siblings, including age at diagnosis, we did not measure the size of their sibship. When studying familial aggregation of disease, the conventional case-control design may lead to potentially biased estimates of the odds ratio due to the lack of control for family size, in particular for extended pedigrees [14]. In our study, analyses restricting family to proband's parents and grandparents were conducted and results were in agreement with the ones including parental and child siblings.

Our results for family history of other diseases provide some reassurance with respect to reporting bias: no excess was reported for these diseases assumed to be unrelated to ALL. In addition, reported prevalence of diabetes, or asthma, in parents was on the low side in both the case and control groups [15,16]. High accuracy in reporting family history of cancer among relatives has been shown, especially for first-degree relatives [17] (who in our study are second-degree relatives to the index child). However, it is still possible that differential recall existed between the case parents and the control parents, even if this study showed underreporting of equal proportions by parents of cases and controls for other factors [18]. Systematic examination of the impact of differential recall suggests that for a risk factor with a low prevalence, the odds ratio is more sensitive to differences in specificity than to differences in sensitivity between the case and the control groups [19]. For family history of cancer, the predictive value of a negative report was found high and non-differential between the case and control groups, resulting in very high specificity in both groups [20].

There are very limited data on family history of cancer among children with ALL; based on these data we cannot rule out the possibility the observation that even rare cancers believed to be sporadic have a genetic contribution.

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